

CHASE: A Corridor-Heuristic Charging Scheduler for Ground and Aerial Vehicles in Wireless Rechargeable Sensor Networks

Liang-Ching Tao¹, Wei-Chen Lin¹ and Pi-Chung Wang^{1,*}

¹ Department of Computer Science and Engineering, National Chung Hsing University, Taichung 402, Taiwan; liangkingtao@gmail.com; H33197@gmail.com

* Correspondence: pcwang.tw@gmail.com

Problem. Wireless rechargeable sensor networks recharge battery-limited nodes in the field [2]. Under dense deployment a ground wireless charging vehicle (WCV) is too slow to meet every deadline, and a pure UAV charger is capped by its battery. Existing hybrid schemes switch from Nearest Job First to Earliest Deadline First only once the former becomes infeasible, so nodes due inside that switch window are already lost [1].

Approach. CHASE unifies both rules inside every WCV leg: it anchors the most urgent target by deadline, then folds in only pending nodes lying within a narrow *bypass corridor* along that leg. We also solve the rendezvous of a UAV returning to a *continuously moving* WCV by a provably contractive fixed-point iteration, which prior hybrid work leaves unspecified.

Results. Over 30 seeds against CHCS [1] and three baselines, the gain is confined to the stressed regime. At 600 and 800 nodes CHASE and CHCS cannot be separated (paired Wilcoxon $p > 0.1$), but from 1000 nodes CHASE sustains **82.8%** survival against CHCS's 74.8% (+8.0 points, $p < 10^{-7}$), peaking at +10.0. Removing the corridor alone costs **19.8** points, locating the effect in the proposed mechanism; the corridor buys them at +48% mileage, yet from 1000 nodes onward CHASE's WCV mileage stays *below* CHCS's (899 vs. 1254 km at 1400 nodes).

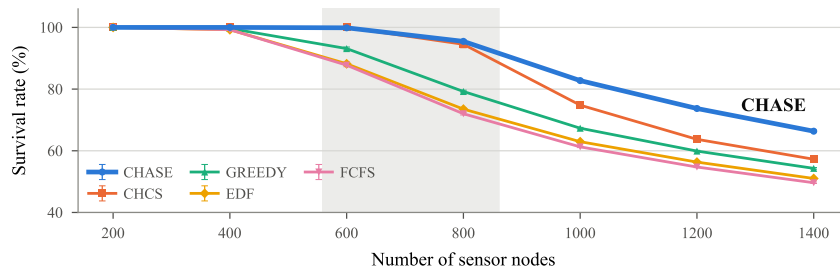


Fig. 1. Survival rate versus node density, 30-seed mean, ± 1 standard error. Shaded: where CHASE and CHCS cannot be separated. The gap to EDF isolates the corridor's effect, since the two share every other component.

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[1] J. Chen *et al.*, "Collaborative hybrid charging scheduling in WRSNs," *IEEE Trans. Veh. Technol.*, 71(8), 8994–9010, 2022.

[2] A. Kaswan *et al.*, "A survey on mobile charging techniques in WRSNs," *IEEE Commun. Surveys Tuts.*, 24(3), 1750–1779, 2022.